

PLASMA SPRAY PROCESSORS

	WORK INSTRUCTION PTA	DOC. REF. NO : QF/QMS/02/ 09 REVISION NO : 00 DATE : 01/03/2019
• Purpose of PTA(plasma transferred arc) is to achieve metallurgical bonding with powder. • Check the dimension of the component and record in the job card. • Check the drawing/ Instruction given by the works manager for thickness of deposition. • Conduct the dye penetration (DPT) test on the component. Clean the component thoroughly after the test. • Mount the component on the machine with suitable support. • Carry out cleaning and degreasing of the component only where required. • Set the parameter as per required standards. • Pre-heat the job upto 200⁰C – 250⁰ C for steel/stainless steel material and 250⁰ C– 300⁰C for stellite material. • Set current on higher end to start 150/190 Amps. (Depending on job thickness and weld deposition). • Reduce current gradually minimum upto 100Amps. (Check dilution and fluidity of weld spot). • For heavy job maintain temperature by continuous heating on non-welding area by burner. Intermediate dimension should be checked as per tolerance limit. • Check the dimension to get the desired thickness of the coating mentioned in the job card. • After removing the component from the machine immerse in vermiculite immediately after welding • Or wrap in Asbestos Cloth for interruption of welding. • Machine or grind the coating area and conduct the DP test and if there are no cracks found continue with machining or grinding to meet the required size. • Check dimensions of the component and record in the job card.		
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